The 30% Ruling and Skilled Migration to the Netherlands

E_EBE2_RP — Research Paper, Period 2.6 (2025–2026)

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1 Introduction

The Dutch 30% ruling (30%-regeling) allows employers to pay up to 30% of a highly skilled migrant's salary tax-free for up to five years (Netherlands Chamber of Commerce, KVK, 2025). It is one of the most generous tax incentives for skilled migrants in Europe and a central instrument in the Netherlands' strategy to attract international talent (Godar, Flamant & Richard, 2021; Vankan et al., 2017). However, the ruling has been subject to repeated reforms: the maximum duration was reduced from eight to five years in 2019, a stepped 30/20/10 structure was briefly introduced in 2024 (later reversed), and a further reduction to 27% is scheduled for 2027 (Bijlsma et al., 2024).

This paper examines the effect of the 2019 reform on skilled migration to the Netherlands. We combine descriptive analysis of migration trends with a fiscal computation of the ruling's budgetary effects, drawing on Eurostat cross-country data, IND permit statistics, and the recent SEO evaluation (Bijlsma et al., 2024). As a complementary exercise, we estimate a difference-in-differences model comparing the Netherlands to peer European countries.

The SEO evaluation (2024), using confidential CBS microdata, found that the 2019 reform reduced inflows of high-earning migrants by 19.6%. Timm, Giuliodori & Muller (2025), studying the earlier 2012 reform, found that expanding eligibility more than doubled migration just above the income threshold, with an elasticity of 1.6–2.7. Our contribution is threefold: (1) we provide an accessible, institutional description of the ruling and its reforms, (2) we compute fiscal effects across salary levels and policy scenarios, connecting the findings from existing studies, and (3) we examine whether the reform effect is detectable in publicly available cross-country data — finding that it is not, due to COVID-19 contamination and data granularity limitations.

2 Institutional Background

2.1 How the 30% Ruling Works

The 30% ruling allows employers to grant up to 30% of a qualifying employee's gross salary as a tax-free allowance, intended to compensate for extraterritorial costs — additional expenses incurred from living outside one's home country. The tax-free portion is not subject to income tax or social security contributions in Box 1. The employee's taxable income is therefore reduced to 70% of gross salary, substantially lowering their effective tax rate (Timm et al., 2025).

To qualify, an employee must be recruited from abroad and possess specific expertise that is scarce on the Dutch labour market. Since 2012, this criterion is operationalised as a minimum salary threshold: €46,660 for 2025 (€35,468 for employees under 30 with a Master's degree) (Netherlands Chamber of Commerce, KVK, 2025). Academic researchers are exempt from the income requirement. The employee must have lived more than 150 km from the Dutch border for the majority of the 24 months prior to starting employment in the Netherlands (Belastingdienst, 2025).

2.2 Timeline of Reforms

The repeated reforms reflect an ongoing tension between the ruling's effectiveness in attracting talent and concerns about its fiscal cost and generosity. As Timm et al. (2025) note, the 2012 reform improved transparency by replacing the subjective "scarce skills" assessment with a clear income threshold, which paradoxically *increased* take-up by making eligibility more predictable for both employers and migrants.

2.3 Who Are the Knowledge Migrants?

The IND/CBS cohort study (Van der Werf et al., 2023) followed two cohorts of non-EEA knowledge migrants who received their first residence permit in 2014 (N=7,764) and 2017 (N=12,192). This provides a unique long-run picture of who comes, how long they stay, and where they work.

```r
# IND/CBS cohort study key statistics (Van der Werf et al., 2023)
# Source: "Kenniswerkers en zoekjaarders in Nederland", IND/CBS, October 2023

# Top nationalities
nationalities <- data.frame(
  Rank = 1:5,
  Nationality = c("Indian", "American", "Chinese", "Turkish/Japanese", "Brazilian/Russian"),
  Share_2014 = c("27%", "13%", "12%", "4-5%", "4%"),
  Share_2017 = c("30%", "12%", "10%", "4-6%", "4-5%")
)
knitr::kable(nationalities, caption = "Top nationalities of knowledge migrants (IND/CBS, 2023)"
```

Table 1: Top nationalities of knowledge migrants (IND/CBS, 2023)

Rank	Nationality	Share_2014	Share_2017
1	Indian	27%	30%
2	American	13%	12%
3	Chinese	12%	10%
4	Turkish/Japanese	4–5%	4–6%
5	Brazilian/Russian	4%	4–5%

Key demographic facts: average age at arrival is 32 years, approximately 70% are male, and over 90% arrive from outside the EU. The dominance of Indian, American, and Chinese nationals reflects the global competition for STEM and business services talent that the 30% ruling is designed to win.

Retention: only 35% of the 2014 cohort was still in the Netherlands at the end of 2020 — six to seven years after arrival. About half (53%) remained after three to four years. Of those who stayed, the vast majority (91%) were employed. Most who left returned to their country of origin (India, USA, China). This retention pattern has important implications for the fiscal analysis, discussed below.

Geographic concentration: knowledge migrants are heavily concentrated in a few cities. Among the 2017 cohort still in the Netherlands at end of 2020: Amsterdam (28%), Eindhoven (8%), Amstelveen (7%), Utrecht (6%), Den Haag (6%). SEO (2024) separately estimates that approximately 5,339 of the 11,168 migrants who came to the Netherlands specifically because of the 30% ruling live in Amsterdam. The two figures are not directly comparable — IND/CBS tracks all knowledge migrants who remained, while SEO's estimate is based on a behavioural model isolating those whose location decision was driven by the tax incentive — but both point to Amsterdam as by far the dominant destination for this group.

Sectors: Among the IND/CBS cohorts, the most common sectors are software development and production (15–16%) and university education (7–12%) (Van der Werf et al., 2023). Timm et al. (2025) report that business services — broadly defined to include IT, legal, consulting, journalism, administration, and financial intermediary services — is by far the largest sector among 30% ruling beneficiaries, accounting for 55.5% of beneficiaries in the post-2012 period, compared to only 15.0% of non-beneficiaries (Table 2.2). This sector also showed the strongest response to the 2012 reform, with an estimated increase of approximately 200% relative to pre-reform levels — larger in absolute terms than all other sectors combined. The IND/CBS software share sits within Timm et al.'s broader business services category. Giarola et al. (2023) report business services (29%) and trade/transport/catering (21%) as the two largest sectors in their sample, which covers a different and broader population of beneficiaries going back to 2002.

Demographics from Giarola et al. (2023): 30% ruling recipients are predominantly male (78%), relatively young (54% under 35), and majority are single (55%) without children (60%). Strikingly, three-fifths have wages placing them in the top 5% of the Dutch wage distribution, and one-third are in the top 1% — confirming that the ruling disproportionately benefits very high earners despite also covering mid-level salaries.

Dialogic (2017) provides more detail on the 2015 user base (N=56,431):

Table 2: User profile and ETK analysis, Dialogic (2017)

Characteristic	Value
Share male	75%
Share higher educated	~90%
Average gross salary (excl. allowance)	€84,000
Median gross salary	€52,000
Share using ruling 5 years	~80%
Real ETK as share of salary (average user)	~29%
Real ETK as share of salary (income >€100k)	<6%
Share: forfait too high (real ETK < 30%)	~50%
Share: forfait appropriate	~35%
Share: forfait too low (real ETK > 30%)	~15%

The skew between average (€84,000) and median (€52,000) salary confirms that a small number of very high earners substantially inflates the average. The ETK analysis is crucial: for incomes above €100,000, real extraterritorial costs fall below 6% of salary, meaning the 30% exemption is predominantly a tax subsidy for high earners rather than a cost reimbursement. This finding explains both the political pressure to reform the ruling and the concentration of behavioural responses among top earners documented by Giarola et al. (2023).

The zoekjaarder pipeline: 84% of job-search-year permit holders in 2017 had previously been in the Netherlands for higher education. Just over half (52–54%) converted to a knowledge worker permit directly after the search year, and 85% found some form of work during or after the search period. This reveals an important pathway: international students → search year → knowledge migrant → potential 30% ruling beneficiary (Van der Werf et al., 2023).

The Netherlands is not alone in offering tax incentives for skilled migrants. Similar schemes exist in Denmark (26% flat rate for up to 7 years), Italy (up to 70% income exemption for returnees), Spain (the "Beckham Law," 24% flat rate for 6 years), and Portugal (the former NHR regime). However, the design varies considerably: some target only researchers, others cap at certain income levels, and durations range from 3 to 10 years. Timm et al. (2025) note that the number of such preferential tax schemes within the EU increased from 5 in the 1990s to 28 by 2021 (Flamant et al., 2021), reflecting intensifying international tax competition for skilled workers.

3 Literature Review

Three studies provide direct evidence on the Dutch 30% ruling, each examining a different reform and margin of adjustment:

Timm, Giuliodori & Muller (2025) study the **2012 reform** (introduction of income threshold). Using CBS administrative microdata and a difference-in-differences design comparing income bins above and below the newly introduced threshold, they find that migration more than doubled in the income range just above the threshold: increases of 376%, 148%, and 108% for migrants within 1%, 5%, and 10% of the threshold respectively. Nearly 11,000 additional migrants arrived between 2012 and 2019 due to the reform. Non-EU migrants were far more responsive (140% increase in the 100–110% range) than EU migrants (48%), and the business services sector saw the largest response (~200%). The implied migration elasticity with respect to the net-of-tax rate ranges from 1.6 to 2.7, higher than Kleven et al.'s (2014) estimate of 1.6 for top earners in Denmark — suggesting mid-level earners are at least as tax-responsive as top earners. Their back-of-the-envelope budget calculations suggest the reform was self-financing: the tax revenue from additional migrants exceeds the cost of the exemption, because beneficiaries still pay taxes on 70% of their income.

Bijlsma et al. / SEO (2024) study the **2019 reform** (duration reduction from 8 to 5 years). Using a similar DiD approach with CBS microdata but comparing income groups above and below the salary threshold, they find a 19.6% decline in inflows for high earners (50–100% above the threshold). The overall migration elasticity is 0.52 (lower than Timm et al. because it captures the response to a less dramatic reform — a duration change rather than an eligibility expansion). The ruling generated €900 million in cumulative net tax revenue over 2016–2022, averaging €128.5 million per year. SEO estimates that complete abolition of the ruling would reduce skilled immigration by approximately 40%.

Giarola, Marie, Cörvers & Schmeets (2023) study the **emigration side** of the 2012 reform. The 2012 reform introduced a 150 km distance criterion, which retroactively removed eligibility from migrants who had arrived from Belgium, Luxembourg, and parts of western Germany and northern France between 2007 and 2011. These workers lost their tax break after 5 years instead of the promised 10. Using Belastingdienst and CBS administrative data on 54,386 unique beneficiaries from 2002–2019, they estimate the causal effect on out-migration decisions using a DiD comparing treated (Belgium/Luxembourg arrivals, 2007–2011) to untreated workers.

Key findings: affected workers stayed **5.3 fewer months (8% less time)** in the Netherlands. The probability of staying beyond 5 years — i.e., past the loss of the tax break — decreased by **13%**. The out-migration elasticity for the top 1% of earners is estimated at **1.48–1.74**, comparable to Kleven et al.'s in-migration estimates. Crucially, workers between the 50th and 95th percentile show **no out-migration response whatsoever** — the effect is driven entirely by the top 5%, and especially the top 1%. Highly mobile individuals (those who had arrived from a third country rather than their origin country) respond more strongly, and family roots (marriage, children) do not significantly reduce the mobility response. Consistent with SEO (2024), the reform is found to be **fiscally cost-neutral**: increased tax receipts from lower-income stayers who remain in the Netherlands offset the revenue lost from departing top earners.

Individual estimates in this literature range from 0.52 (Bijlsma et al., 2024) to 1.74 (Giarola et al., 2023), with Kleven et al. (2014) and Akcigit et al. (2016) finding elasticities around 1.6 for top earners.

Vankan et al. / Dialogic (2017) provide the official evaluation commissioned by the Ministry of Finance that directly triggered the 2019 reform. Using CBS microdata on 56,431 users in 2015 and a survey of 1,463 beneficiaries, they find the ruling effective on all three stated objectives. Their key empirical finding — that **approximately 80% of users apply the ruling for five years or fewer** — became the direct justification for the 2019 duration reduction. They estimate 1,765–5,575 additional foreign workers in the Netherlands due to the ruling (of whom 1,000–4,000 earn below €100k). On the ETK question, they find real extraterritorial costs average 29% of salary for a typical user, but only 20% across the full user base because high earners have proportionally lower costs — at incomes above €100k, real ETK fall below 6% of salary, making the ruling effectively a tax subsidy for high earners. No evidence of displacement of Dutch workers was found.

4 Data: 30% Ruling Usage

4.1 SEO Evaluation Data

The most comprehensive data on 30% ruling usage comes from the 2024 SEO evaluation "Kunde, Kosten en Keuzes" (Bijlsma et al., 2024), which used CBS administrative microdata. We digitise their key tables for our analysis.

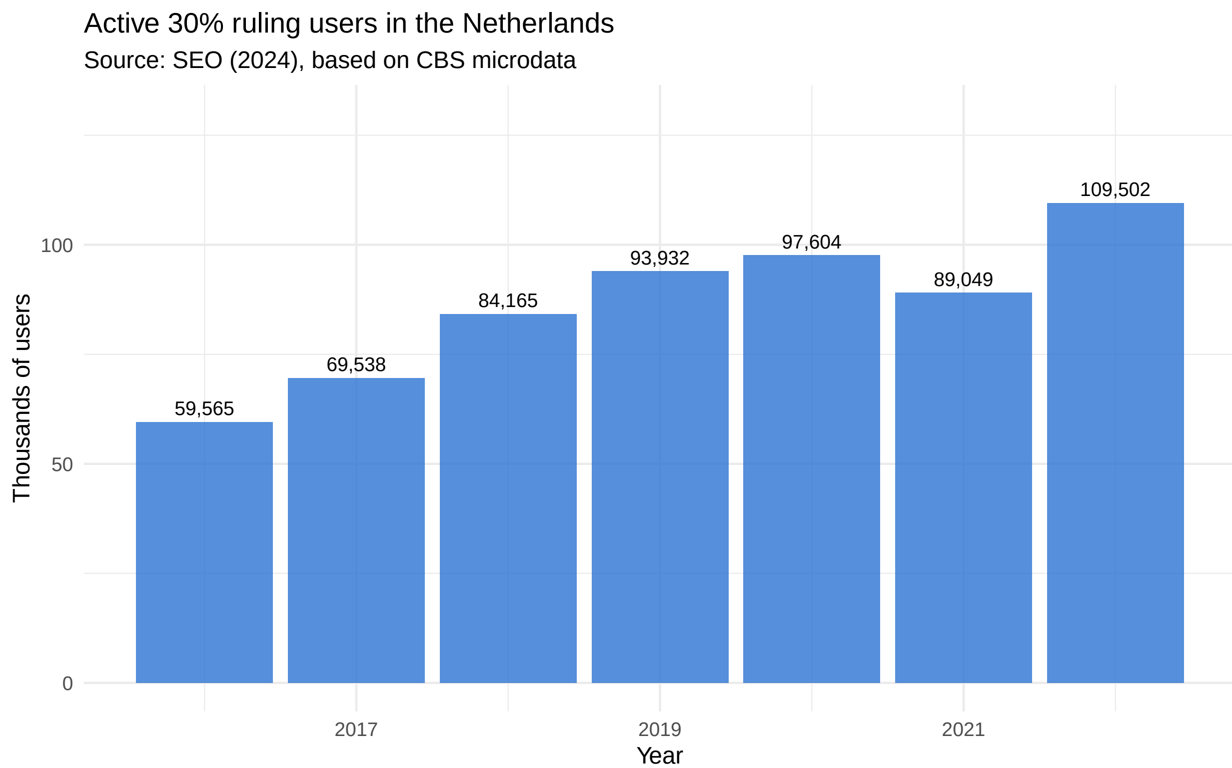
```r
# Active users and fiscal cost per year (SEO 2024, Figure 3.1 / Table 3.11)
# Source: Bijlsma et al. (2024), o.b.v. CBS Microdata
seo_users <- data.frame(
  year = 2016:2022,
  active_users = c(59565, 69538, 84165, 93932, 97604, 89049, 109502),
  total_income_m = c(4597, 5401, 6273, 7265, 8103, 6686, 8001),
  tax_exempt_m = c(1379, 1620, 1882, 2179, 2431, 2006, 2400),
  tax_revenue_m = c(1287, 1512, 1756, 2034, 2269, 1872, 2240),
  foregone_revenue_m = c(552, 648, 753, 872, 972, 802, 960),
  net_revenue_m = c(736, 864, 1004, 1162, 1296, 1070, 1280)
)

seo_users
```

```
##   year active_users total_income_m tax_exempt_m tax_revenue_m
## 1 2016        59565           4597         1379          1287
## 2 2017        69538           5401         1620          1512
## 3 2018        84165           6273         1882          1756
## 4 2019        93932           7265         2179          2034
## 5 2020        97604           8103         2431          2269
## 6 2021        89049           6686         2006          1872
## 7 2022       109502           8001         2400          2240
##   foregone_revenue_m net_revenue_m
## 1                552           736
## 2                648           864
## 3                753          1004
## 4                872          1162
## 5                972          1296
## 6                802          1070
## 7                960          1280
```

The number of active 30% ruling users nearly doubled from 59,565 in 2016 to 109,502 in 2022. This growth tracks the broader international trend of rising labour migration to the EU, but also reflects the Netherlands' strong labour market and the attractiveness of the ruling itself. Total income of beneficiaries reached €8 billion in 2022, of which €2.4 billion was tax-exempt under the ruling.

```r
ggplot(seo_users, aes(x = year, y = active_users / 1000)) +
  geom_col(fill = "#2C73D2", alpha = 0.8) +
  geom_text(aes(label = comma(active_users)), vjust = -0.5, size = 3) +
  labs(title = "Active 30% ruling users in the Netherlands",
       subtitle = "Source: SEO (2024), based on CBS microdata",
       x = "Year", y = "Thousands of users") +
  scale_y_continuous(limits = c(0, 130)) +
  theme_minimal()
```



4.2 New Inflows by Cohort

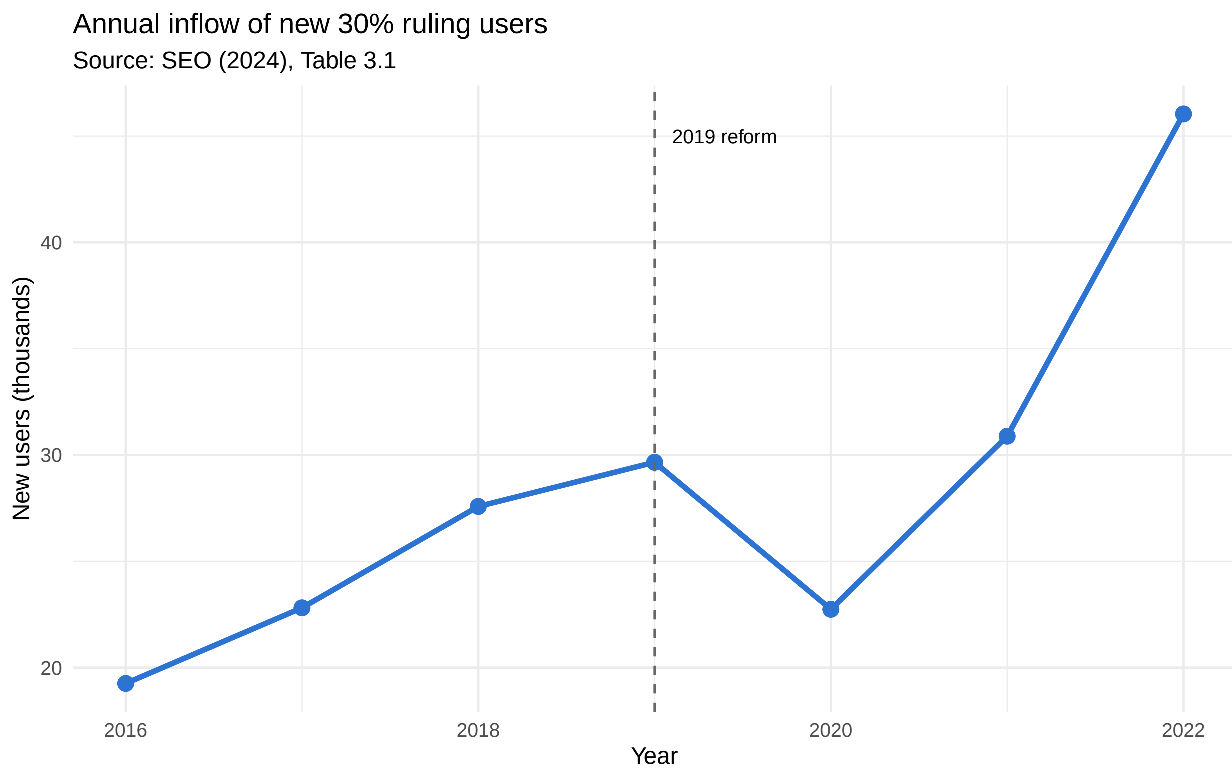
Annual inflow of new 30% ruling users (SEO 2024, Table 3.1)

```
##   year new_users
## 1 2016     19254
## 2 2017     22811
## 3 2018     27579
## 4 2019     29658
## 5 2020     22742
## 6 2021     30884
## 7 2022     46042
```

New inflows grew steadily from 19,254 in 2016 to 29,658 in 2019, dipped to 22,742 in 2020 (COVID), and surged to 46,042 in 2022 — likely reflecting post-pandemic backlog clearing. The 2020 dip is notable: it coincides with both the COVID travel restrictions and the first full year under the shortened 5-year duration, making it impossible to disentangle the two effects in aggregate data.

```
ggplot(seo_inflow, aes(x = year, y = new_users / 1000)) +
  geom_line(linewidth = 1.2, color = "#2C73D2") +
  geom_point(size = 3, color = "#2C73D2") +
  geom_vline(xintercept = 2019, linetype = "dashed", color = "grey40") +
  annotate("text", x = 2019.1, y = 45, label = "2019 reform", hjust = 0, size = 3) +
  labs(title = "Annual inflow of new 30% ruling users",
```

```r
    subtitle = "Source: SEO (2024), Table 3.1",
    x = "Year", y = "New users (thousands)") +
  theme_minimal()
```



4.3 Belastingdienst Application Data (Woo Request)

30% ruling applications — from Woo (freedom of information) request, September 2024 Source: Besluit Woo-verzoek over de 30%-regeling, Belastingdienst

```
##   year applications approved rejected approval_rate
## 1 2018        30753    25963     1078          96.0
## 2 2019        33376    34209     1615          95.5
## 3 2020        27629    29730     1348          95.7
## 4 2021        33143    30162     1291          95.9
## 5 2022        48414    42467     1586          96.4
## 6 2023        44463    48163     1883          96.2
```

The Belastingdienst data confirms the pattern: applications dipped in 2020 (COVID) but surged in 2022–2023. The approval rate has remained consistently above 93%, suggesting that the eligibility criteria are well understood by employers and that few ineligible applications are filed.

4.4 Migration Elasticities in the Literature

The responsiveness of migrants to tax incentives is a central parameter for policy evaluation. The table below compares key elasticity estimates from the literature:

Table 3: Migration elasticity estimates with respect to net-of-tax rate

Study	Country	Reform	Population	Elasticity
Kleven et al. (2014)	Denmark	Foreigners' tax scheme	Top earners (in-migration)	1.60
Timm et al. (2025) — lower bound	Netherlands	2012 income threshold	Mid-level earners (in-migration)	1.60
Timm et al. (2025) — upper bound	Netherlands	2012 income threshold	Mid-level earners (in-migration)	2.70
Bijlsma et al./SEO (2024)	Netherlands	2019 duration reduction	All beneficiaries (in-migration)	0.52
Giarola et al. (2023) — top 1%	Netherlands	2012 distance criterion (emigration)	Top 1% earners (out-migration)	1.61
Moretti & Wilson (2017)	United States	State-level R&D tax credits	Star scientists	1.80

The SEO elasticity of 0.52 is notably lower because it captures the response to a *duration* change (8→5 years) rather than an eligibility change. A shorter benefit period reduces the present value of the tax advantage but does not eliminate it, so the behavioural response is smaller than to a binary eligibility change like the 2012 reform studied by Timm et al.

5 Data: Cross-Country Skilled Migration (Eurostat)

```r
# Eurostat first residence permits by occupation (migr_resocc)
data <- read.csv("data/1_migr_resocc__custom_21740522_linear.csv")

# Clean: keep relevant columns, remove EU aggregates, rename
data <- data %>%
  select(-DATAFLOW, -LAST.UPDATE, -freq, -duration, -citizen, -unit, -OBS_FLAG, -CONF_STATUS)
data <- data[!grepl("European Union", data$geo), ]
colnames(data) <- c("reason", "country", "year", "permits")
```

```r
# The Netherlands barely uses the EU Blue Card - relies on its own kennismigrant scheme
data[data$country == "Netherlands" & data$reason == "EU Blue card" & data$year >= 2015,
     c("year", "permits")]
```

```
##     year permits
## 954 2015      20
```

```
## 955 2016      42
## 956 2017      58
## 957 2018      78
## 958 2019     132
## 959 2020     141
## 960 2021     147
## 961 2022     216
## 962 2023     203
## 963 2024     250
```

The Netherlands issued only 250 EU Blue Cards in 2024, compared to over 13,000 "Highly skilled workers" permits. This confirms that the Netherlands relies almost entirely on its national knowledge migrant scheme, making Dutch-specific policy changes (like the 30% ruling reform) disproportionately important for skilled migration flows.

```r
# Focus on "Highly skilled workers" permits for NL + control countries
# Controls: Denmark, Sweden, Austria (full data, no concurrent policy changes)
# Excluded: Germany (missing 2019-2022), Belgium & France (data reclassification issues)
analysis <- data[data$reason == "Highly skilled workers" &
                 data$country %in% c("Netherlands", "Denmark", "Sweden", "Austria"), ]
analysis <- analysis[analysis$year >= 2015 & analysis$year <= 2024 & analysis$year != 2019, ]
```

5.1 Cross-Country Comparison

```r
analysis[c("country", "year", "permits")] %>%
  pivot_wider(names_from = country, values_from = permits)
```

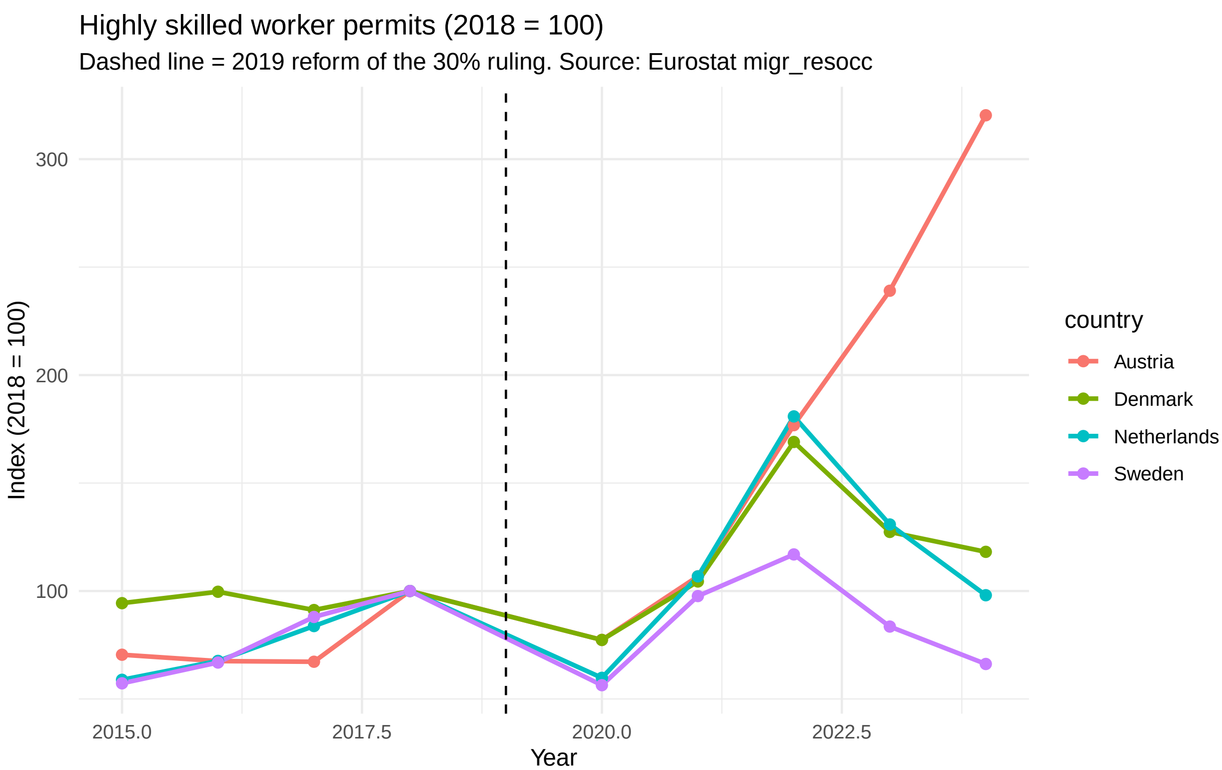
```
## # A tibble: 9 x 5
##    year Austria Denmark Netherlands Sweden
##   <int>   <int>   <int>       <int>  <int>
## 1  2015    1173    5457        7909   4527
## 2  2016    1124    5762        9084   5288
## 3  2017    1119    5273       11252   6957
## 4  2018    1664    5782       13432   7907
## 5  2020    1287    4472        8031   4459
## 6  2021    1777    6037       14341   7722
## 7  2022    2942    9773       24293   9247
## 8  2023    3978    7364       17569   6605
## 9  2024    5330    6831       13170   5235
```

Belgium and France were excluded as controls due to data quality concerns: Belgium's "Highly skilled workers" permits dropped from 3,555 to 20 in a single year (2016–2017), almost certainly reflecting a change in reporting methodology rather than actual migration. Germany — conceptually the strongest control (no competing tax scheme) — is missing data for 2019–2022 in this category.

5.2 Indexed Trends Plot

```
plot_data <- analysis %>%
  group_by(country) %>%
  mutate(index = permits / permits[year == 2018] * 100) %>%
  ungroup()

ggplot(plot_data, aes(x = year, y = index, color = country)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_vline(xintercept = 2019, linetype = "dashed") +
  labs(title = "Highly skilled worker permits (2018 = 100)",
       subtitle = "Dashed line = 2019 reform of the 30% ruling. Source: Eurostat migr_resocc",
       x = "Year", y = "Index (2018 = 100)") +
  theme_minimal()
```



The indexed plot shows roughly parallel pre-reform trends for the Netherlands, Denmark, and Sweden (supporting the parallel trends assumption). After 2019, all countries experience a COVID dip and subsequent recovery. The Netherlands shows the sharpest spike in 2022 (~180% of 2018 levels), followed by a steep decline. Austria diverges dramatically, likely due to country-specific factors unrelated to the Dutch reform.

6 Fiscal Computation

This section quantifies the budgetary effects of the 30% ruling for both the government (foregone tax revenue) and individual expats (tax savings), following the supervisor's guidance. We compute effects across salary levels and compare three policy regimes.

6.1 Tax Function (2025 Brackets)

```r
# Netherlands Box 1 income tax brackets for 2025
# Source: KVK / Belastingdienst
calc_tax <- function(income) {
  if (income <= 0) return(0)
  b1 <- 38441; b2 <- 76817
  r1 <- 0.3582; r2 <- 0.3748; r3 <- 0.4950
  if (income <= b1) return(income * r1)
  if (income <= b2) return(b1 * r1 + (income - b1) * r2)
  return(b1 * r1 + (b2 - b1) * r2 + (income - b2) * r3)
}
```

6.2 Per-Person Tax Savings

```r
salaries <- seq(50000, 200000, by = 10000)

fiscal <- data.frame(
  gross_salary = salaries,
  tax_normal = sapply(salaries, calc_tax),
  tax_30pct = sapply(salaries, function(x) calc_tax(x * 0.70)),
  tax_27pct = sapply(salaries, function(x) calc_tax(x * 0.73))
)

fiscal$saving_30 <- fiscal$tax_normal - fiscal$tax_30pct
fiscal$saving_27 <- fiscal$tax_normal - fiscal$tax_27pct
fiscal$eff_rate_normal <- round(fiscal$tax_normal / fiscal$gross_salary * 100, 1)
fiscal$eff_rate_30 <- round(fiscal$tax_30pct / fiscal$gross_salary * 100, 1)
fiscal$eff_rate_27 <- round(fiscal$tax_27pct / fiscal$gross_salary * 100, 1)

fiscal[, c("gross_salary", "saving_30", "saving_27",
           "eff_rate_normal", "eff_rate_30", "eff_rate_27")]
```

```
##    gross_salary saving_30 saving_27 eff_rate_normal eff_rate_30 eff_rate_27
## 1         50000  5564.879  5027.579            36.2        25.1        26.1
## 2         60000  6746.400  6071.760            36.4        25.2        26.3
## 3         70000  7870.800  7083.720            36.6        25.3        26.4
## 4         80000  9377.797  8478.277            37.2        25.4        26.6
```

```
## 5           90000 11704.197 10692.237            38.5         25.5         26.7
## 6          100000 14030.597 12906.197            39.6         25.6         26.7
## 7          110000 16335.000 14701.500            40.5         25.7         27.2
## 8          120000 17820.000 16038.000            41.3         26.4         27.9
## 9          130000 19305.000 17374.500            41.9         27.1         28.5
## 10         140000 20790.000 18711.000            42.4         27.6         29.1
## 11         150000 22275.000 20047.500            42.9         28.1         29.6
## 12         160000 23760.000 21384.000            43.3         28.5         30.0
## 13         170000 25245.000 22720.500            43.7         28.8         30.3
## 14         180000 26730.000 24057.000            44.0         29.2         30.7
## 15         190000 28215.000 25393.500            44.3         29.5         30.9
## 16         200000 29700.000 26730.000            44.6         29.7         31.2
```

The annual tax saving ranges from roughly €5,500 at €50,000 to €30,000 at €200,000. The ruling reduces effective tax rates dramatically: for an expat earning €100,000, the effective rate drops from 39.6% to 25.6%. The SEO report (2024) found a mean salary of €83,000 among beneficiaries, placing the typical annual saving at approximately €9,000–€10,000.

An important nuance from the IND/CBS cohort study (Van der Werf et al., 2023): only 35% of the 2014 cohort was still in the Netherlands after 6–7 years. This means that for the majority of beneficiaries, the reduction in duration from 8 to 5 years does not actually bite — they would have left anyway before reaching year 6. The reform's main effect is therefore concentrated on the most highly committed migrants: those who planned to stay beyond 5 years.

This is confirmed by Giarola et al.'s (2023) finding that only the top 5% — and especially the top 1% — show significant out-migration responses to the loss of the tax break, while workers between the 50th and 95th percentile show no behavioural change at all. High earners are more likely to be long-stayers who planned to use the full duration, which is precisely why the duration reduction hits them hardest.

The Dialogic (2017) ETK analysis provides an additional lens: for incomes above €100,000, real extraterritorial costs fall below 6% of salary. This means the 30% exemption for top earners is largely a tax subsidy rather than a cost reimbursement — which simultaneously explains why the ruling is politically contested and why high earners respond so strongly to changes in its generosity. Crucially, Dialogic also found that only ~5% of Dutch employers use the ruling, but just 171 firms (0.05% of all employers) account for 43% of all users. This concentration in multinational headquarters and large tech/finance firms has important implications: the ruling's impact on the investment climate is real but disproportionately benefits large internationally mobile firms, which have the most to gain from relocating if the fiscal environment deteriorates.

```r
fiscal_long <- fiscal %>%
  select(gross_salary, saving_30, saving_27) %>%
  pivot_longer(cols = c(saving_30, saving_27),
               names_to = "regime", values_to = "saving") %>%
  mutate(regime = ifelse(regime == "saving_30",
                         "30% ruling (current)", "27% ruling (from 2027)"))

ggplot(fiscal_long, aes(x = gross_salary, y = saving, color = regime)) +
  geom_line(linewidth = 1.2) +
```

```r
  geom_point(size = 2) +
  scale_x_continuous(labels = comma_format(prefix = "€")) +
  scale_y_continuous(labels = comma_format(prefix = "€")) +
  labs(title = "Annual tax saving by salary level",
       subtitle = "Comparison of 30% and 27% regime (2025 brackets)",
       x = "Gross annual salary", y = "Annual tax saving (€)", color = "Regime") +
  theme_minimal()
```

6.3 Lifetime Benefit: Three Policy Scenarios

```r
discount_rate <- 0.03

pv <- function(annual_saving, years, r = discount_rate) {
  sum(annual_saving / (1 + r)^(1:years))
}

fiscal$pv_pre2019 <- sapply(fiscal$saving_30, function(s) pv(s, 8))
fiscal$pv_post2019 <- sapply(fiscal$saving_30, function(s) pv(s, 5))
fiscal$pv_post2027 <- sapply(fiscal$saving_27, function(s) pv(s, 5))
fiscal$reduction_2019 <- round((1 - fiscal$pv_post2019 / fiscal$pv_pre2019) * 100, 1)
fiscal$reduction_2027 <- round((1 - fiscal$pv_post2027 / fiscal$pv_pre2019) * 100, 1)
```

```r
fiscal[, c("gross_salary", "pv_pre2019", "pv_post2019", "pv_post2027",
           "reduction_2019", "reduction_2027")]
```

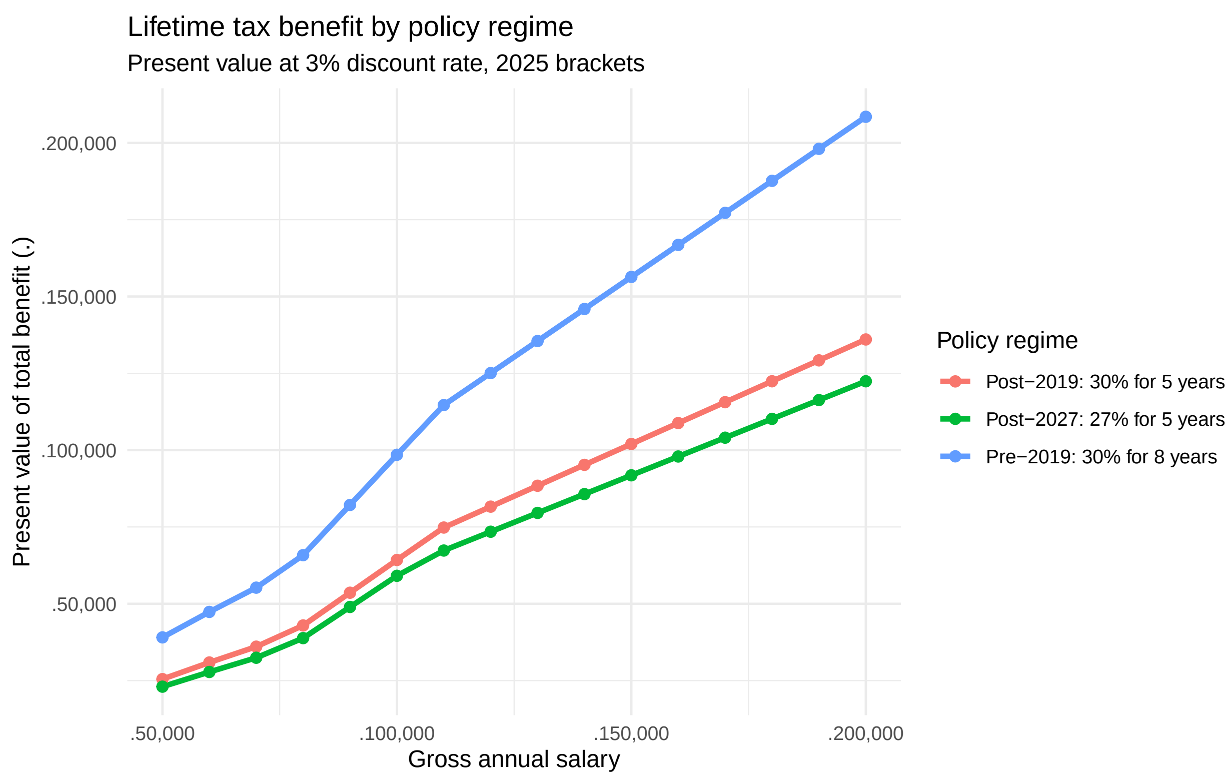
```
##    gross_salary pv_pre2019 pv_post2019 pv_post2027 reduction_2019
## 1         50000   39063.74    25485.52    23024.84           34.8
## 2         60000   47357.65    30896.54    27806.88           34.8
## 3         70000   55250.59    36045.96    32441.36           34.8
## 4         80000   65829.25    42947.56    38828.02           34.8
## 5         90000   82159.86    53601.79    48967.31           34.8
## 6        100000   98490.47    64256.02    59106.60           34.8
## 7        110000  114666.67    74809.52    67328.57           34.8
## 8        120000  125090.91    81610.38    73449.34           34.8
## 9        130000  135515.16    88411.25    79570.12           34.8
## 10       140000  145939.40    95212.11    85690.90           34.8
## 11       150000  156363.64   102012.98    91811.68           34.8
## 12       160000  166787.89   108813.84    97932.46           34.8
## 13       170000  177212.13   115614.71   104053.24           34.8
## 14       180000  187636.37   122415.57   110174.02           34.8
## 15       190000  198060.62   129216.44   116294.79           34.8
## 16       200000  208484.86   136017.30   122415.57           34.8
##    reduction_2027
## 1            41.1
## 2            41.3
## 3            41.3
## 4            41.0
## 5            40.4
## 6            40.0
## 7            41.3
## 8            41.3
## 9            41.3
## 10           41.3
## 11           41.3
## 12           41.3
## 13           41.3
## 14           41.3
## 15           41.3
## 16           41.3
```

The 2019 reform cut the lifetime tax benefit by approximately 34.8% across all salary levels. For an expat earning €100,000, this means a reduction from ~€98,500 to ~€64,300 — a loss of over €34,000 in present value. The 2027 reform adds a further reduction, bringing total lifetime benefit down by ~41% relative to the pre-2019 regime.

```r
pv_long <- fiscal %>%
  select(gross_salary, pv_pre2019, pv_post2019, pv_post2027) %>%
  pivot_longer(cols = starts_with("pv_"), names_to = "regime", values_to = "pv") %>%
```

```r
  mutate(regime = case_when(
    regime == "pv_pre2019" ~ "Pre-2019: 30% for 8 years",
    regime == "pv_post2019" ~ "Post-2019: 30% for 5 years",
    regime == "pv_post2027" ~ "Post-2027: 27% for 5 years"))

ggplot(pv_long, aes(x = gross_salary, y = pv, color = regime)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  scale_x_continuous(labels = comma_format(prefix = "€")) +
  scale_y_continuous(labels = comma_format(prefix = "€")) +
  labs(title = "Lifetime tax benefit by policy regime",
       subtitle = "Present value at 3% discount rate, 2025 brackets",
       x = "Gross annual salary", y = "Present value of total benefit (€)",
       color = "Policy regime") +
  theme_minimal()
```



6.4 Aggregate Fiscal Cost and Cross-Check with SEO

SEO (2024) reports: ~110,000 users, ~€1 billion fiscal cost, avg salary €83,000

```r
# Our calculation:
representative_salary <- 83000
representative_saving <- calc_tax(representative_salary) - calc_tax(representative_salary * 0.
n_users_2022 <- 109502
```

```r
cat("Per-person annual saving (€83k salary): €",
    format(round(representative_saving), big.mark = ","), "\n")
```

```
## Per-person annual saving (€83k salary): € 10,076
```

```r
cat("Bottom-up fiscal cost estimate: €",
    format(round(representative_saving * n_users_2022 / 1e9, 2)), " billion\n")
```

```
## Bottom-up fiscal cost estimate: € 1.1  billion
```

```r
cat("SEO (2024) reported fiscal cost: ~€1.0 billion (2022)\n")
```

```
## SEO (2024) reported fiscal cost: ~€1.0 billion (2022)
```

```r
cat("SEO (2024) net revenue (method 2): €57.2 million (2022)\n\n")
```

```
## SEO (2024) net revenue (method 2): €57.2 million (2022)
```

```r
# Per-person lifetime benefit reduction from 2019 reform
pv_saving <- pv(representative_saving, 8) - pv(representative_saving, 5)
cat("Per-person lifetime benefit reduction (8→5 years): €",
    format(round(pv_saving), big.mark = ","), "\n")
```

```
## Per-person lifetime benefit reduction (8→5 years): € 24,585
```

```r
cat("Reduction as percentage: ",
    round((1 - pv(representative_saving, 5) / pv(representative_saving, 8)) * 100, 1),
    "%\n", sep = "")
```

```
## Reduction as percentage: 34.8%
```

Our bottom-up estimate of the fiscal cost is somewhat below SEO's reported €1 billion because our calculation uses 2025 tax brackets and doesn't account for the full salary distribution (SEO's figure includes all income levels, weighted by actual frequency). The key insight from SEO's second method is that the ruling generated a cumulative €900 million in net revenue over 2016–2022 — meaning it pays for itself when accounting for the tax revenue from migrants who would not have come without the incentive.

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